Data types ÷

Data type is an instruction to the compiler. It tells the computer two critical things÷

1. What Kind of data is being stored

2. How much memory space is to be set.

variable

Example ÷ char m ;

Datatype

Here m is an variable in which char type data will be store

variable

Example ÷ int age ;

Datatype

Again here age is an int type variable in which int type value will be store

int age = 2 ; ✓

int age = "Gaurav"; ✗

DATA Types :-

Primary data type	Derived data type	user defined data type
Pre-defined data type.	Data types derived from the primitive or built-in data types are called Derived Data types.	
Char int float double void	array function pointer structure union	Typedef enum

Data Type	Qualifier
Char	signed, Unsigned
int	Short, long, signed, unsigned
float	No qualifier
double	long
void	No qualifier

signed → both +ve & -ve
unsigned → only +ve

Data Type	Compiler 16-bit size (bytes)	32-bit size (bytes)	64-bit size (bytes)
Char or Signed char	1	1	1
Unsigned char	1	1	1
Short int	1	2	2
unsigned Short int	1	2	2
int	2	4	4
unsigned int	2	4	4
long int	4	4	8
unsigned long int	4	4	8
Float	4	4	4
Double	8	8	8
long Double	10	16	16
Pointer	4	4	8

Ordinary variable का size उसके data type पर depend करता है।

Pointer variable का size उसके data type पर depend नही करता